

Dispensing Platform Development Roadmap

Objective

Compare three implementation paths for obtaining an additional dispensing platform and increasing automation capability while considering the limited remaining project duration (~2 months), limited available engineering effort, and company-purchase constraints.

Overview of the Three Stages

Stage 1 – MK3S Replica

Replicate the existing dispensing platform using the current architecture:

- Used Prusa MK3S / MK3S+ motion platform
- Vacuum chuck
- BLTouch surface mapping
- Hamilton syringe dispensing
- Syringe drive assembly
- Klipper
- Raspberry Pi Zero W / Zero WH
- Existing CAD files
- Existing software workflow

Goal

Obtain a second operational machine as quickly as possible with minimal engineering effort.

Stage 2 – MK3S Replica with Automation Upgrades

Build a second MK3S-based platform and implement two additional automation features:

Automatic Needle Homing

- Soft-contact homing station
- Automatic needle reference acquisition
- Removal of manual Z-offset adjustment
- Improved repeatability
- Designed to protect fragile Hamilton syringe needle

Estimated engineering effort: **~1 day**

Servo-Controlled Lid Handling

- Servo-actuated disposable container lid

- Automated opening and closing of containers
- Fully scriptable operation through Klipper
- Reduced manual pauses between dispensing cycles

Estimated engineering effort: ~1 day

Goal

Reduce operator involvement while maintaining the proven MK3S architecture.

Stage 3 – Dedicated Laboratory Dispensing Platform

Purpose-built dispensing machine featuring:

- Industrial linear modules
- Ballscrew-driven axes
- Dedicated Z-axis
- Dedicated syringe axis
- Manta M8P controller or equivalent
- CB1 / Raspberry Pi compute module
- TMC2209 or TMC5160 drivers
- Industrial frame
- Automatic needle homing
- Automated lid handling
- Optional camera support

Goal

Develop a long-term laboratory platform optimized for rigidity, maintainability, automation, and future expansion.

Feature Comparison

Feature	Stage 1: MK3S Replica	Stage 2: MK3S + Automation	Stage 3: Dedicated Lab System
Vacuum Chuck	✓	✓	✓
Hamilton Syringe	✓	✓	✓
BLTouch Surface Mapping	✓	✓	Optional
Klipper	✓	✓	✓
Existing CAD Files	✓	✓	X
Existing Firmware Basis	✓	✓	X

Feature	Stage 1: MK3S Replica	Stage 2: MK3S + Automation	Stage 3: Dedicated Lab System
Camera Support	X	X	Optional / future addition
Automatic Needle Homing	X	✓	✓
Servo Lid Automation	X	✓	✓
Fully Scripted Workflow	Partial	Mostly / full	Full
Operator Intervention	High	Low	Very Low
Industrial Linear Modules	X	X	✓
Ballscrew Axes for XYZ	X	X	✓
Dedicated Syringe Axis	Custom / required	Custom / required	✓
Expandability	Medium	High	Very High
Mechanical Rigidity	Medium	Medium	High

Cost Comparison

Item	Stage 1	Stage 2	Stage 3
Hardware Cost (€)	650–1,050	800–1,300	1,500–2,500+
Additional Engineering Time	3–7 days	5–9 days	4–8 weeks
Technical Risk	Low	Low–Medium	Medium–High
Procurement Complexity	Medium	Medium	High
Company-Purchase Friendliness	Medium	Medium	High

Note: Stage 1 and Stage 2 depend on sourcing a used MK3S / MK3S+ unless an existing spare printer is available. Used machines are available, but company purchasing may be less straightforward if bought from private sellers.

Automation Benefits

Automatic Needle Homing

Current workflow:

1. Mesh PET surface.

2. Manually move needle.
3. Manually establish needle reference.
4. Start dispensing.

Stage 2 workflow:

1. Mesh PET surface.
2. Move to homing station.
3. Automatically acquire needle reference.
4. Start dispensing.

Benefits:

- Improved repeatability
- Reduced setup time
- Reduced operator dependence
- Lower risk of damaging the fragile Hamilton syringe needle during manual referencing

Servo Lid Automation

Current workflow:

1. Open sample container manually.
2. Dispense material.
3. Close container manually.
4. Resume or stop software manually.

Stage 2 workflow:

1. Open container automatically.
2. Dispense material.
3. Close container automatically.
4. Continue scripted workflow.

Benefits:

- Reduced operator intervention
- Fewer manual stops between samples
- Fully scriptable dispensing sequence
- Improved process consistency

Decision Matrix

Criterion	Weight	Stage 1	Stage 2	Stage 3
Initial Cost	25%	5	4	2
Development Time	25%	5	4	1

Criterion	Weight	Stage 1	Stage 2	Stage 3
Technical Risk	20%	5	4	2
Automation Capability	10%	2	5	5
Maintainability	10%	3	3	5
Mechanical Performance	10%	3	3	5
Weighted Assessment		High	Very High	Medium

Recommendation

Considering:

- Approximately two months of remaining project time
- Development is not a full-time activity
- Existing MK3S platform is already validated
- Existing CAD and software infrastructure are available
- Needle homing and servo automation are expected to require approximately one day each
- A fully dedicated lab system may not be completed and validated before the end of the project period

Recommended Option

Stage 2 – MK3S Replica with Automation Upgrades

This option provides:

- A second operational platform
- Low technical risk
- Moderate cost
- Significant reduction in operator involvement
- Automatic needle referencing
- Automated container handling
- Better scripted workflow automation

while avoiding the cost, procurement effort, and development risk associated with a complete machine redesign.

Long-Term Recommendation

If the dispensing platform becomes a permanent laboratory asset beyond the current project, Stage 3 should be pursued as a future development effort rather than as part of the current project timeline.

Stage 2 Bill of Materials

This BOM assumes a second MK3S / MK3S+ base is purchased and converted into the automated dispensing platform.

Prices are partially approximated and should be treated as planning values.

Stage 2 BOM Summary

Category	Estimated Cost (€)
MK3S / MK3S+ base machine	250–550
Vacuum chuck system	120
Control electronics and compute	20
BLTouch / probing	20
Syringe and dispensing hardware	120–300
Syringe driver axis	120–250
Needle homing station	25–75
Servo lid automation	25–80
Wiring, cable sleeves, cable chain, connectors	50
Mechanical hardware and printed parts	40–100
Contingency	100–150
Estimated total	890–1715 €

This assumes the MK3S / MK3S+ can be sourced at a reasonable used price and that some existing printed parts / design files can be reused.

Procurement Notes

MK3S / MK3S+ Availability

Used MK3S / MK3S+ machines are currently available on the German used market, commonly in the range of approximately **180–450 €**, depending on condition and accessories.

However, this is not ideal for company purchase because many listings are private sales without standard company invoicing.

For a company/university purchase, preferred options are:

1. Buy from a commercial refurbisher or reseller if available.
2. Buy through a marketplace that supports business invoices.
3. Internally justify a private used purchase if accounting allows it.
4. As a fallback, purchase a newer Prusa platform, but this may reduce compatibility with existing MK3S CAD files and conversion parts.

Syringe Driver Axis

The syringe driver should not be treated as a minor accessory. It requires:

- Stepper motor
- Leadscrew or ballscrew
- Linear guide
- Plunger carriage
- Endstop or reference position
- Mechanical syringe clamp
- Driver electronics or use of an available MK3S motor channel

Planning cost: **120–250 €**

Needle Homing

The automatic needle homing station is low-cost but high-value.

Recommended design:

- Soft conductive touch pad
- Spring/flexure compliance
- High-value resistor for current limiting
- Very slow probing speed
- Immediate retract after trigger

Planning cost: **25–75 €**

Servo Lid Automation

Servo lid automation is also low-cost but high-value.

Recommended design:

- Metal gear servo
- Printed or machined linkage
- Mechanical overtravel allowance
- Optional open/closed confirmation switch

Planning cost: **25–80 €**

Conclusion

Stage 2 remains the preferred option.

It keeps the validated MK3S-based approach while adding the two automation features that provide the highest immediate practical benefit:

1. Automatic needle homing
2. Servo-controlled container lid handling

The expected total budget for a complete Stage 2 platform is approximately:

890–1,715 €